

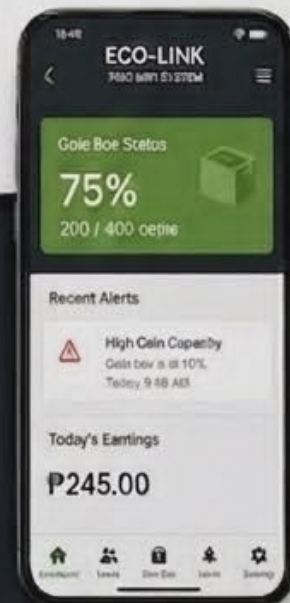
Eco-Link

A SOLAR-POWERED PISO WIFI SYSTEM
with Integrated Coin Capacity Alert

SPMP

SOFTWARE PROJECT MANAGEMENT PLAN

ASTRYX



Capstone Project Eco-Link: A Solar-powered Piso WiFi System with Integrated Coin Capacity Alert

Capstone Project Eco-Link: A Solar-powered Piso WiFi System with Integrated Coin Capacity Alert

for

Eco link: A Solar-powered Piso WiFi SSystem
with Integrated Coin Capacity Alert

Capstone Project*Training Management Endeavors (TrainME):**Web-based Training Needs Assessment and***Change History**

Version	Primary Author(s)	Description of Version	Date Completed
Draft 1	Wenda D. Panes	First rough draft.	7/1/2017
Draft 2	Wenda D. Panes	Changed Adviser's name, changed project due date, fixed misspelling.	7/8/2017
Version 1	Wenda D. Panes	Changed presentation due date	8/25/2017

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1. Overview

The **Eco-Link** project focuses on developing an advanced, sustainable Piso WiFi vending machine integrated with solar energy and Internet of Things (IoT) capabilities. The system is designed to address the operational challenges of traditional vending machines, such as the need for constant manual monitoring and dependence on grid power.

1.1. Project Summary

1.1.1.

Purpose, scope and objectives

1. Purpose and Scope

The primary goal is to create a unified solution that integrates renewable energy harvesting with a real-time **coin capacity alert feature**. The system allows owners to manage multiple units remotely by receiving mobile notifications when a coin holder is full, thereby reducing maintenance costs and operational downtime.

The project encompasses several critical domains:

- **Solar Power Integration:** Utilizing photovoltaic power to ensure energy autonomy in remote or off-grid locations.
- **IoT Monitoring:** Implementing microcontrollers (e.g., Arduino, ESP8266, or Raspberry Pi) to manage power routing, telemetry, and system diagnostics.
- **Automated Alerts:** Developing a communication architecture to push real-time status updates and coin capacity notifications to web-based dashboards or mobile devices.

2. Technical Objectives

The research establishes several technical foundations for the project:

- **Power Resilience:** Using rule-based charge controllers to monitor solar output and battery health, preventing hardware wear and deep discharging.
- **Smart Vending Logic:** Transitioning from reactive maintenance to a proactive model through integer-based error-coding and self-diagnostic capabilities.
- **Network Security:** Implementing robust authentication via RADIUS and LDAP, alongside secure captive portals to protect user data and prevent unauthorized access.
- **Data Management:** Utilizing time-series databases (like InfluxDB) and visualization tools (Grafana) to handle high-density telemetry data across multiple distributed units.

3. Ethical and Operational Considerations

To ensure a successful deployment, the project must navigate various constraints and

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responsibilities:

- **Inclusivity:** Maintaining a hybrid approach that supports both traditional coin payments and digital mobile payments to remain accessible to all demographics.
- **Compliance:** Adhering to the **Data Privacy Act of 2012** to ensure responsible management of user information.
- **Sustainability:** Reducing carbon emissions and overhead energy costs through off-grid solar harvesting.
- **Challenges:** Mitigating risks related to weather-dependent power unreliability, cybersecurity threats, and limited internet connectivity in rural area

1.1.2.

Assumptions and constraints

4.1 Project Assumptions

The project development and deployment are based on the following underlying assumptions derived from existing literature and system requirements:

- **Energy Availability:** It is assumed that the deployment site receives sufficient solar irradiance to allow the photovoltaic panels to harvest enough ambient energy for autonomous operation.
- **Hardware Compatibility:** The system assumes that low-cost microcontrollers (e.g., ESP8266 or Arduino) can successfully interface with both the solar charge controllers and the physical coin-slot mechanisms.
- **Network Availability:** It is assumed that a minimum level of Wi-Fi or GSM connectivity is available at the installation site to facilitate the transmission of real-time telemetry and mobile alerts.
- **User Adoption:** There is an assumption that the target demographic will be comfortable using a hybrid payment system that includes both traditional coins and digital mobile gateways.
- **Security Resilience:** It is assumed that implementing layered Multi-Factor Authentication (MFA) and RADIUS protocols will effectively render standard brute-force attacks mathematically infeasible.

4.2 Project Constraints

The development and operational effectiveness of the Eco-Link system are limited by the following constraints:

- **Environmental Limitations:** The reliability of the power source is strictly constrained by weather conditions, which can lead to energy unreliability during prolonged periods of low sunlight.
- **Battery Lifespan:** Traditional battery technology has a finite lifespan, which acts as a primary constraint on the long-term maintenance costs and network uptime in off-grid environments.
- **Connectivity Gaps:** Real-time monitoring and emergency notifications are constrained by the potential for limited or unstable internet connectivity in remote areas.
- **Mobile Operating System Constraints:** Secure passwordless authentication (FIDO2) may be hindered by "mini-browser" limitations in certain mobile operating systems that lack necessary API compatibility.
- **Economic Factors:** Scalability for small-scale deployments is significantly restricted by cost constraints and the initial investment required for high-quality hardware.

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- **Regulatory Compliance:** The system must operate within the legal boundaries
 - set by the Data Privacy Act of 2012 regarding the collection and management of user data.
- **Hardware Maintenance:** Physical components are susceptible to wear and improper user behavior, requiring localized intervention despite remote monitoring capabilities.

1.1.3.

Project deliverables

5.1 Software Deliverables: Eco-Link System (Fully Functional Version)

- The primary software deliverable of this project is the fully integrated Eco-Link system designed to provide renewable-energy-powered internet access with intelligent monitoring, secure authentication, and real-time telemetry capabilities. The completed software package includes the following major components:

Embedded Firmware Module

- The embedded firmware serves as the core control logic of the Eco-Link system. It operates on microcontroller platforms such as the Raspberry Pi or Arduino and is responsible for managing power distribution, solar energy monitoring, and automated source switching. The firmware implements hysteresis-based source selection algorithms to minimize unnecessary relay switching and reduce hardware degradation.

Captive Portal and Authentication System

- The project includes a secure captive portal environment that allows users to access the wireless network through a controlled pre-authentication interface. The system incorporates layered authentication mechanisms, SSL/TLS certificate validation, and secure session handling to reduce the risk of unauthorized access and Man-in-the-Middle (MITM) attacks. The portal supports both traditional coin-based access and digital payment gateway integration.

IoT Telemetry and Data Transmission Pipeline

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- An integrated Internet of Things (IoT) telemetry framework is implemented to facilitate the collection and transmission of system performance data. Local nodes communicate operational metrics, including voltage levels, energy utilization, and connectivity status, through Wi-Fi or GSM communication channels to a centralized
- database infrastructure such as InfluxDB.

Cloud-Based Monitoring Dashboard

- The system includes a cloud-managed monitoring dashboard utilizing platforms such as Grafana. This dashboard enables administrators and system owners to remotely monitor operational parameters, including battery voltage, solar charging performance, coin-slot capacity, network activity, and active power-source status in real time.

Automated Alert and Diagnostic Module

- An automated notification subsystem is integrated into the Eco-Link platform to improve preventive maintenance and operational reliability. The module continuously evaluates system conditions and detects anomalies such as undervoltage, overvoltage, power instability, and hardware malfunction. When abnormal conditions are identified, the system generates corresponding error codes and transmits maintenance alerts to the administrator through configured communication channels.
-

5.2 Project Documentation Deliverables

- Comprehensive project documentation shall be provided to support the development, deployment, maintenance, and future enhancement of the Eco-Link system. These documents ensure proper software lifecycle management and technical reference availability.

Software Requirements Specification (SRS)

- The Software Requirements Specification document defines the complete functional and non-functional requirements of the Eco-Link system. Functional requirements include hybrid payment support, coin-capacity monitoring, telemetry reporting, and automated alert generation. Non-functional requirements address system security, reliability, scalability, performance efficiency, and compliance with

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secure authentication protocols such as RADIUS and LDAP.

Software Project Management Plan (SPMP)

- The Software Project Management Plan outlines the overall project framework, including development methodology, implementation timeline, resource allocation, risk management procedures, and deployment strategies. The document also addresses sustainability objectives, renewable-energy integration, and compliance with legal and ethical standards, including the Data Privacy Act of 2012.

Software Design Document (SDD)

- The Software Design Document provides the technical architecture and detailed design specifications of the Eco-Link system. It includes system diagrams, software stack descriptions, communication workflows, and hardware-software integration processes for Real-Time Linux and embedded microcontroller platforms. The document also describes the telemetry data management model and the mechanisms used to maintain data consistency across distributed monitoring nodes.

Software Test Documentation (STD)

- The Software Test Documentation contains the complete testing procedures, validation methods, and testing results conducted throughout system development. This includes penetration testing procedures, authentication security validation, certificate verification testing, sensor calibration results, and power-failure response evaluation. Security assessments may involve tools such as Hashcat to evaluate authentication resilience against brute-force attacks.

User Guide and Installation Manual

- The User Guide and Installation Manual provide detailed instructions for deploying, configuring, operating, and maintaining the Eco-Link system. The manual includes procedures for installing solar hardware components, configuring IoT communication modules, accessing the remote monitoring dashboard, interpreting system alerts, and performing troubleshooting and preventive maintenance operations based on the system's structured error-coding framework.

1.1.4.

Project Schedule

The following timeline outlines the development phases, from initial research and system design to final documentation and system evaluation.

Phase	Activity Description	Estimated Duration
Phase 1: Planning	Research of existing technologies, system design, and literature review.	2 Months
Phase 2: Development	Prototyping, mechatronic construction, and laboratory testing of solar/IoT modules.	2 Months
Phase 3: Integration	Field deployment, captive portal security testing, and awareness campaigns.	2 Months
Phase 4: Evaluation	Impact assessment, system resilience evaluation, and operational monitoring.	2 Months
Phase 5: Completion	Final documentation (SRS, SDD, STD), reporting, and project dissemination.	1 Months

1.2. Budget Summary

The estimated budget covers the core hardware components required to build a sustainable, IoT-integrated solar vending unit.

Hardware & Materials

Item Category	Component Examples	Estimated Cost (PHP)
Power System	Photovoltaic panels, rule-based charge controller, batteries.	₱15,000 - ₱25,000
Computing Units	Raspberry Pi or Arduino with ESP8266 Wi-Fi module.	₱3,500 - ₱6,000
Vending Logic	Mechatronic coin slots, internal robotics, freezer/oven units.	₱5,000 - ₱10,000
IoT/Sensors	Voltage sensors, current sensors, and load-disconnection relays.	₱2,000 - ₱4,000
Casing & Misc.	Rugged housing, wiring, and thermal management materials.	₱3,000 - ₱5,000

2. References

[1] R. Stephens, *Beginning Software Engineering*, IN: John Wiley & Sons, Inc., 2015.

[2] F. Tsui and O. Karam, *Essentials of Software Engineering*, MA: Jones and Bartlett Publishers, 2011.

Websites

[3] IEEE Std 830-1998 (Revision of IEEE Std 830-1993)

[4] IEEE Std 1058-1998 (Revision and redesignation of IEEE Std 1058.1-1987, incorporating IEEE Std 1058-1998 and IEEE Std 1058a-1998)

<https://prod.teamgantt.com/gantt>

3. Definitions

To properly understand the **Eco-Link SPMP**, key technical terms and acronyms are defined as reflected in Table 3.

Table 3. Definitions

Term	Definition
Eco-Link	The official project name for the Solar-Powered Piso WiFi System with Integrated Coin Capacity Alert.
FIDO2	A set of standards for passwordless authentication used in modern network security.
HCI	Human-Computer Interaction; the study of how people interact with computers and to what extent computers are or are not developed for successful interaction with human beings.
IoT	Internet of Things; a network of interconnected smart devices capable of collecting and exchanging data.
LDAP	Lightweight Directory Access Protocol; used for centralized identity management and authentication.
MFA	Multi-Factor Authentication; a layered security approach requiring multiple forms of verification.
MitM	Man-in-the-Middle; a cyberattack where a malicious actor secretly intercepts and relays communications between two parties.
PACELC	A theorem used in distributed data stores to evaluate system behavior during network partitions and normal operations.
PV	Photovoltaic; a method of generating electric power by using solar cells to convert energy from the sun.
RADIUS	Remote Authentication Dial-In User Service; a networking protocol that

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	provides centralized Authentication, Authorization, and Accounting (AAA) management.
RBAC	Role-Based Access Control; an approach to restricting system access to authorized users based on their roles.
SDD	Software Design Document; a written description of a software product that a software designer writes to give a software development team an overall guidance of the architecture.
SPMP	Software Project Management Plan; the document that outlines the deliverables, schedule, and budget for the project.
SRS	Software Requirements Specification; a document that describes what the software will do and how it will be expected to perform.
SSL/TLS	Secure Sockets Layer / Transport Layer Security; cryptographic protocols designed to provide communications security over a computer network.
STD	Software Test Document; a document that describes the testing scope and activities to ensure the software meets its requirements.

4. Project Organization

This section describes interfaces to organizational entities external to the **Eco-Link** project; identifies the internal organizational structure of **Eco-Link**; and defines the roles and responsibilities of each unit involved in the **Eco-Link** project.

4.1. External structure

Eco-Link is developed to address the need for sustainable, remotely managed internet kiosks in off-grid or resource-limited environments. The system interfaces with renewable energy providers and local administrative units to ensure operational resilience and compliance with the **Data Privacy Act of 2012**. The External Organization Structure Chart is presented in Figure 1.

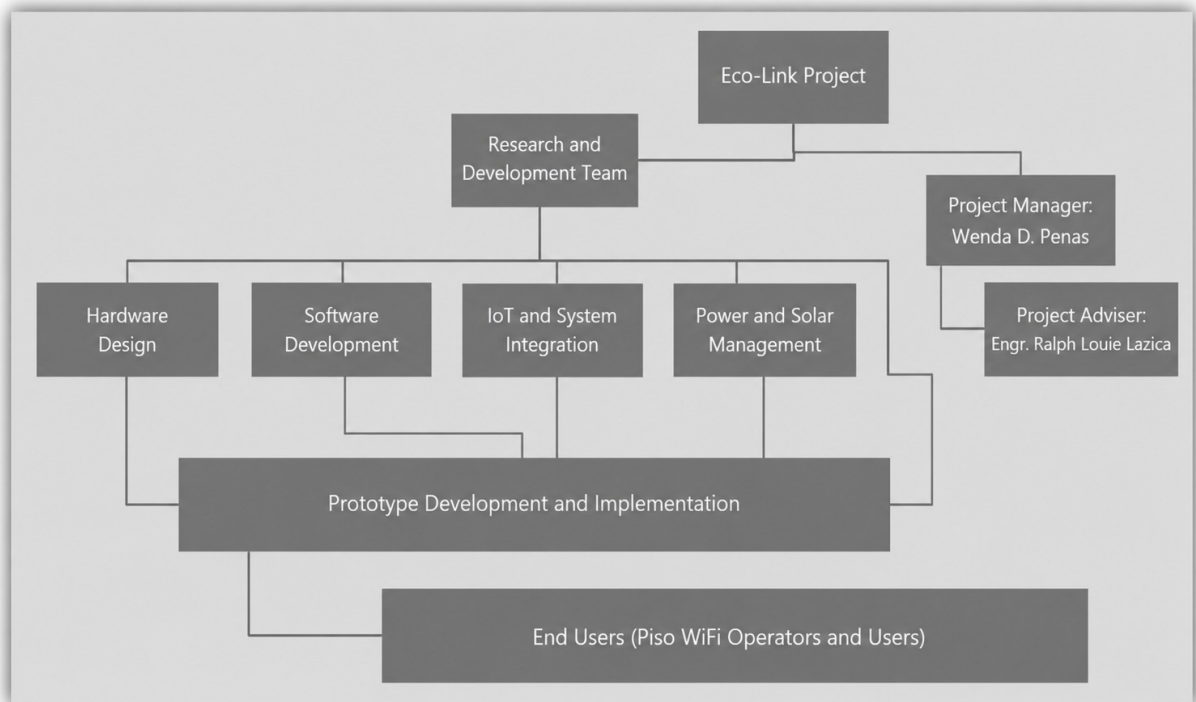


Figure 1 – Eco-link External Organizational Structure

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4.2. Internal structure

The internal organizational structure of **Eco-Link** is designed to facilitate parallel development across hardware, software, and data management units. This ensures that energy autonomy and network security are integrated seamlessly into the final vending solution.

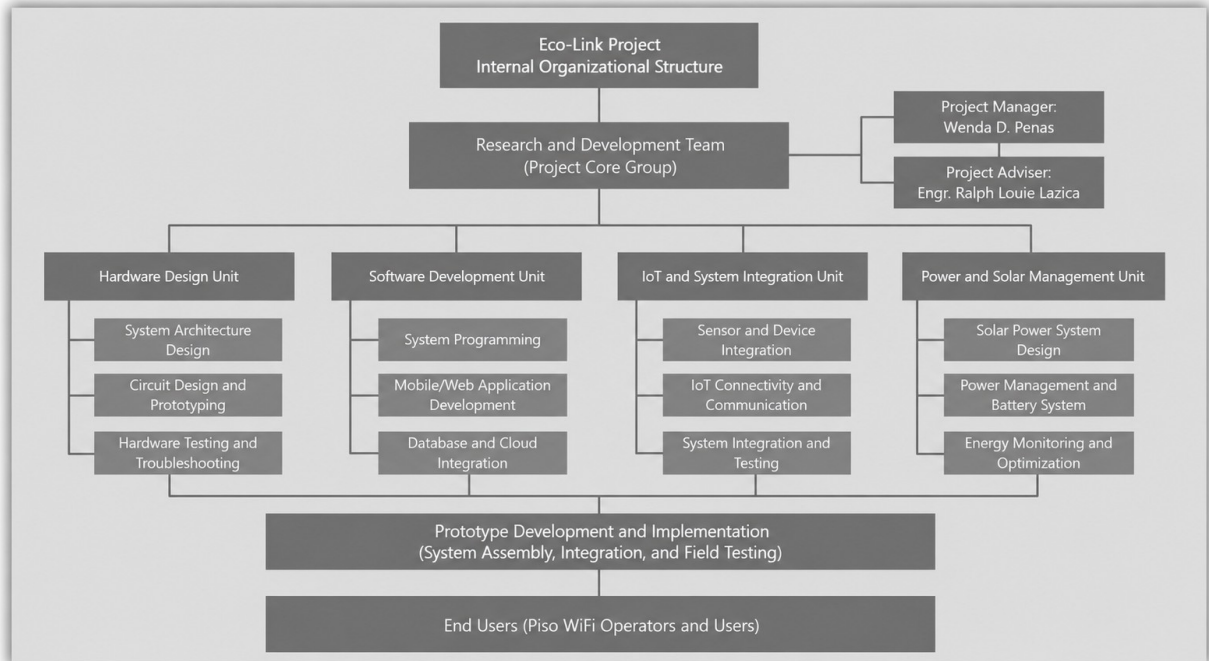


Figure 2. Eco-Link Internal Structure

4.3. Roles and responsibilities

The **Eco-Link** project is developed by a single researcher who assumes multiple specialized roles to ensure the integration of solar power, IoT, and network security. Table 4 depicts the matrix of these roles and their respective responsibilities.

Table 4. Eco-Link Project Team Roles and Responsibilities

Role	Responsibilities
Project Manager	— Responsible for the overall planning, scheduling, and budget management of the Eco-Link system. .
System Architect	— Designs the unified hardware and software architecture, ensuring the integration of solar energy with IoT modules.
Hardware Engineer	— Handles the mechatronic construction, photovoltaic panel installation, and the integration of coin-slot sensors.
Software Developer	— Develops the embedded logic for power management, the captive portal interface, and the automated mobile notification system.
Security Analyst	— Implements layered authentication (RADIUS/LDAP) and ensures the system complies with the Data Privacy Act of 2012 .
Data Administrator	— Manages the telemetry pipelines and centralized dashboards (InfluxDB and Grafana) for real-time monitoring.

5. Managerial process plans

This clause of the SPMP specifies the **Eco-Link** management processes for the project. It includes the project start-up plan, risk management plan, project work plan, project control plan, and project closeout plan.

5.1. Start-up plan

The **Eco-Link** start-up plan specifies the estimation plan, staffing plan, resource acquisition plan, and training plan. These plans may be incorporated directly or by reference to other plans. .

5.1.1. Staffing plan

The **Eco-Link** Staff Plan depicted in Table 5 shows the name, affiliation to the project as well as the status of staff involved in the project.

Table 5. TrainME Staff Plan

Name	Affiliation to project	Status since July 2017	Email Address
Wend D. a Panes	Project Manager	Full Time	wendapanes@gmail.com
Wend D. a Panes	System Analyst	Full Time	wendapanes@gmail.com
Wend D. a Panes	Developer/ Programmer	Full Time	wendapanes@gmail.com
Wend D. a Panes	QA	Full Time	wendapanes@gmail.com
Wend D. a Panes	Document Controller	Full Time	wendapanes@gmail.com
Engr. Ralph P. Laviste	Adviser	Full Time	rlaviste@gmail.com

5.1.3.

Resource acquisition plan

For the duration of **Eco-Link** project development, hiring of additional human resource is not necessary. The Project manager who will assume multi-roles will be the sole responsible to deliver both software and documents requirement. Non-human resources however are needed. The description and dates required for each resource are reflected in Table 6.

Table 6. Eco-Link Non-human resource acquisition plan

Resources	Description
Solar Power Hardware	Photovoltaic panels and energy harvesting modules are essential for enabling sustainable, autonomous operation.
Embedded Controllers	Switches, wireless routers, and internet facility both in the laboratory and faculty room will be utilized by the developer as scheduled.
Intelligent Controllers	Raspberry Pi units will be acquired to serve as rule-based charge controllers for managing real-time power routing.
IoT Connectivity Modules	ESP8266 or ESP32 microcontrollers are needed to bridge local hardware with remote management systems.
Mechatronic Units	Coin-operated modules and sensors must be procured to facilitate the Piso WiFi vending logic and coin capacity alerts.
Data Repository	A centralized InfluxDB time-series database is required to store high-fidelity telemetry data for visualization.
Security Framework	RADIUS and LDAP architectures are necessary to implement layered security and hardware-level filtering.

5.1.4.

Project work plan

The work plan for **Eco-Link** specifies the work activities, schedule, and resource allocations necessary to produce a functional solar-powered WiFi system with coin capacity alerts.

5.2.1. Work activities

The project is divided into several major work packages to ensure the systematic integration of hardware and software.

- **WP 1: Hardware Prototyping & Energy Integration** – Focuses on assembling the photovoltaic panels, Raspberry Pi charge controllers, and battery storage.
- **WP 2: Mechatronic & Vending Logic** – Involves the installation of the coin-operated modules and the programming of the coin capacity sensors.
- **WP 3: IoT & Telemetry Development** – Setup of the ESP8266/ESP32 nodes and establishing the data pipeline to InfluxDB.
- **WP 4: Security & Authentication Layer** – Implementation of the RADIUS server, LDAP filtering, and WebAuthn API for the captive portal.
- **WP 5: System Testing & Validation** – Includes energy resilience testing (checking for deep discharge prevention) and alert accuracy testing.

5.2.2. Schedule allocation

The project follows a phased timeline over the course of the semester/year. Key milestones include:

1. **Hardware Assembly Completion:** Integration of solar components and controllers.
2. **Beta Firmware Release:** Deployment of the mechatronic logic and local WiFi vending.
3. **Cloud Dashboard Launch:** Completion of the Grafana telemetry visualization.
4. **Final System Acceptance:** Successful end-to-end test of coin alerts and solar autonomy.

5.2.3. Resource allocation

Resources are allocated based on the priority of hardware stability. The Project Manager will dedicate 40% of the timeline to Hardware/Energy integration, 30% to Software/IoT development, and 30% to Security and Document Control.

5.3. Project control plan

This clause specifies the mechanisms for monitoring the **Eco-Link** project's progress and

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ensuring it meets its research objectives.

5.3.1. Requirements control plan

To ensure the system meets its sustainable goals, requirements are tracked through a traceability matrix. Any changes to the hardware specifications (e.g., switching solar panel wattage) must be documented to assess the impact on energy harvesting capacity.

5.3.2. Quality control plan

Quality is monitored through specific technical metrics:

- **Uptime Monitoring:** Measuring the system's ability to remain powered during low-light periods.
- **Alert Latency:** Testing the speed of mobile notifications when the coin holder reaches 90% capacity.
- **Authentication Success:** Validating that the passwordless WebAuthn login works across different mobile "mini-browsers."

5.3.3. Reporting plan

Progress reports will be submitted to the Adviser on a bi-weekly basis. These reports will highlight completed work packages, energy harvesting data logs, and any technical bottlenecks encountered in the IoT communication pipeline.

This section describes in detail the decomposition of the total scope of work of the **Eco-Link** project to be carried out by the project team to accomplish the project objectives and create the required deliverables.

5.2.1. Work activities

The **Eco-Link** Work Breakdown Structure (WBS) is designed to organize work activities and their relationships. Work activities are decomposed to a level that allows accurate estimate of resource requirements and schedule duration for each activity. The **Eco-Link** WBS Top-Down Map View is presented in Figure 3. Activity listings and deliverables for each phase are shown in succeeding figures.

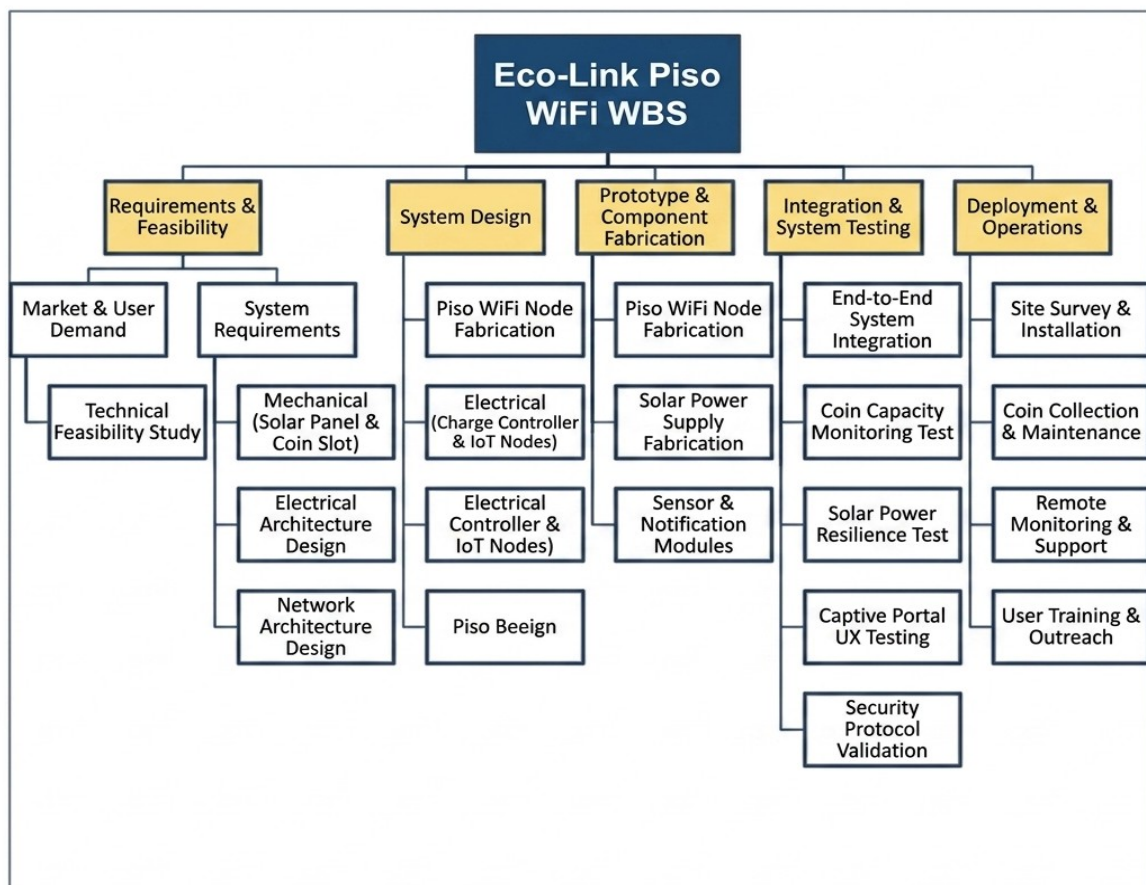


Figure 3. Eco-Link General Work Breakdown Structure

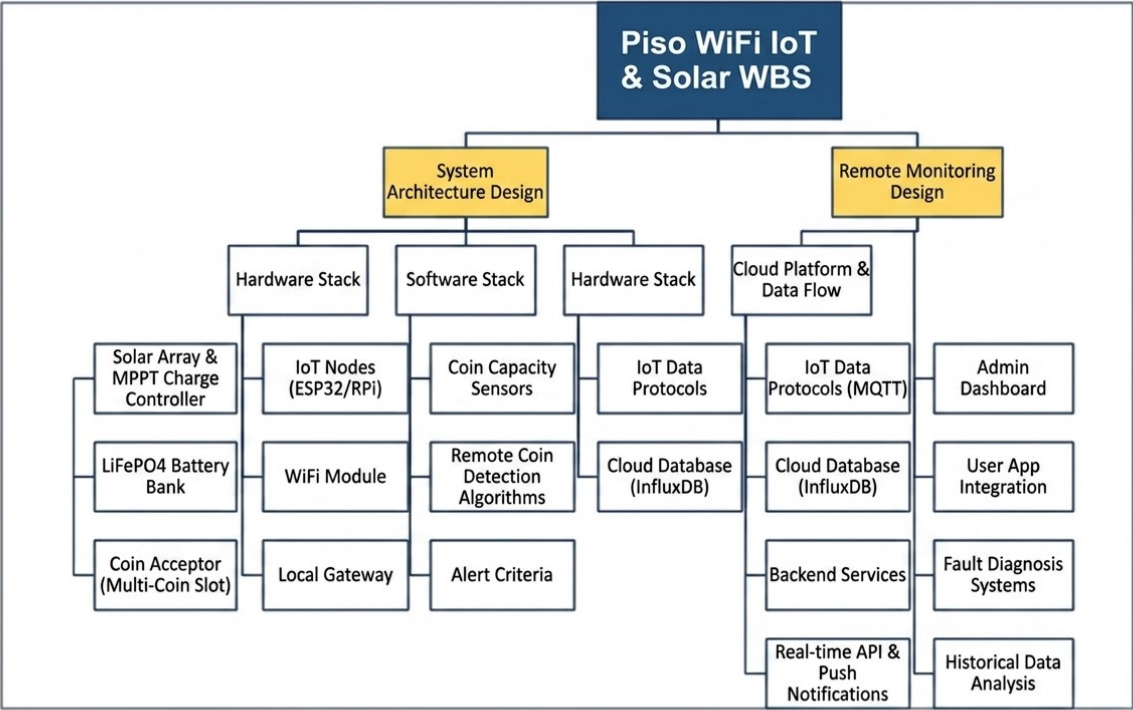


Figure 4. Requirements Analysis WBS

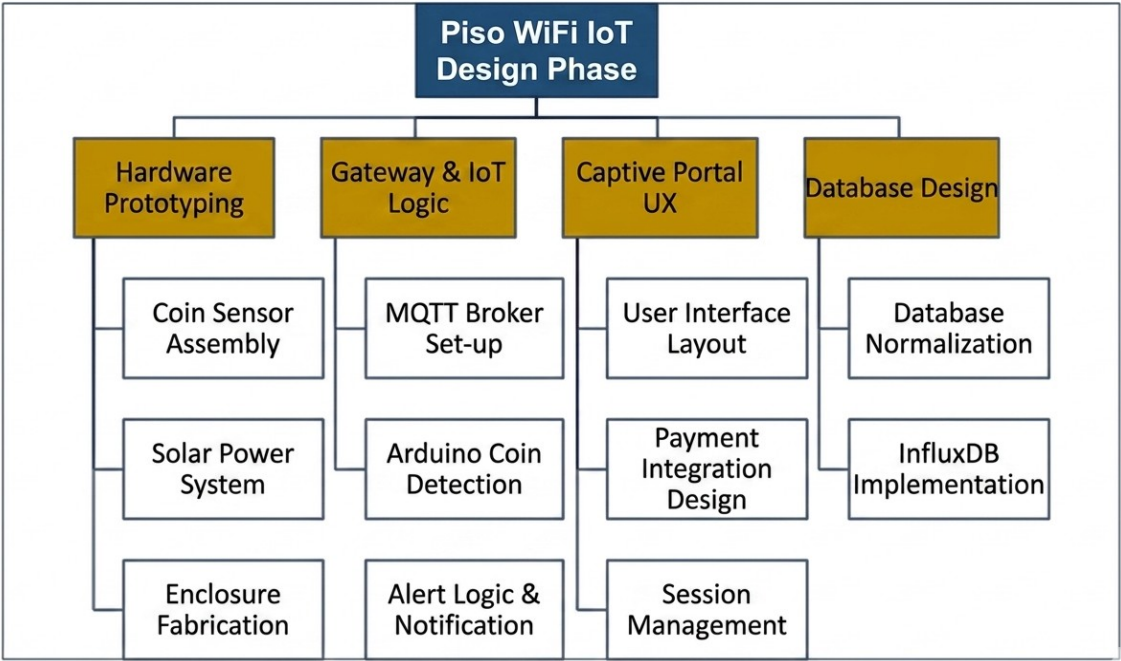
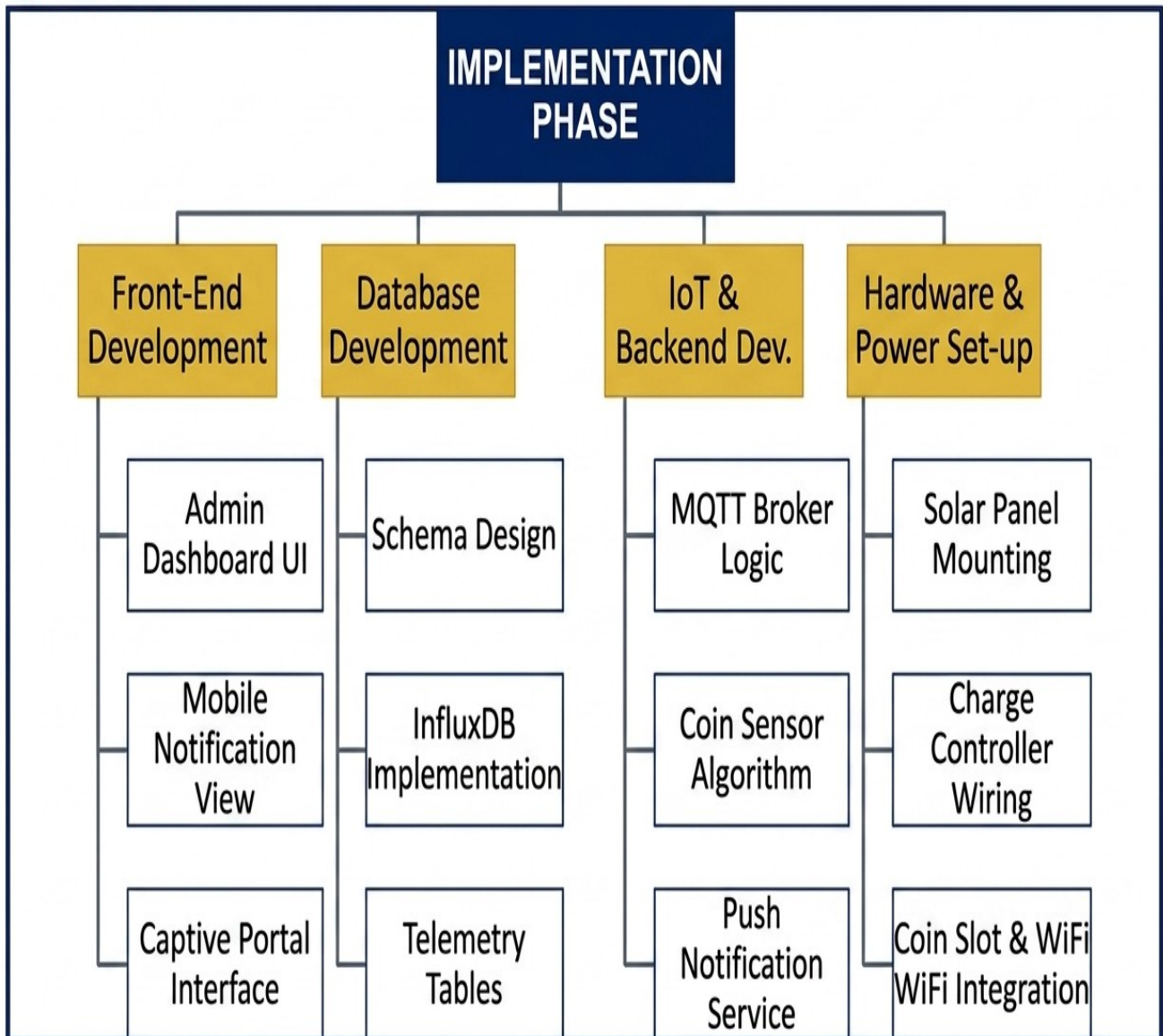


Figure 5. Piso wifi iot design WBS

Figure 6 Eco-Link Piso wifi. Implementation WBS

Eco-Link Piso WiFi



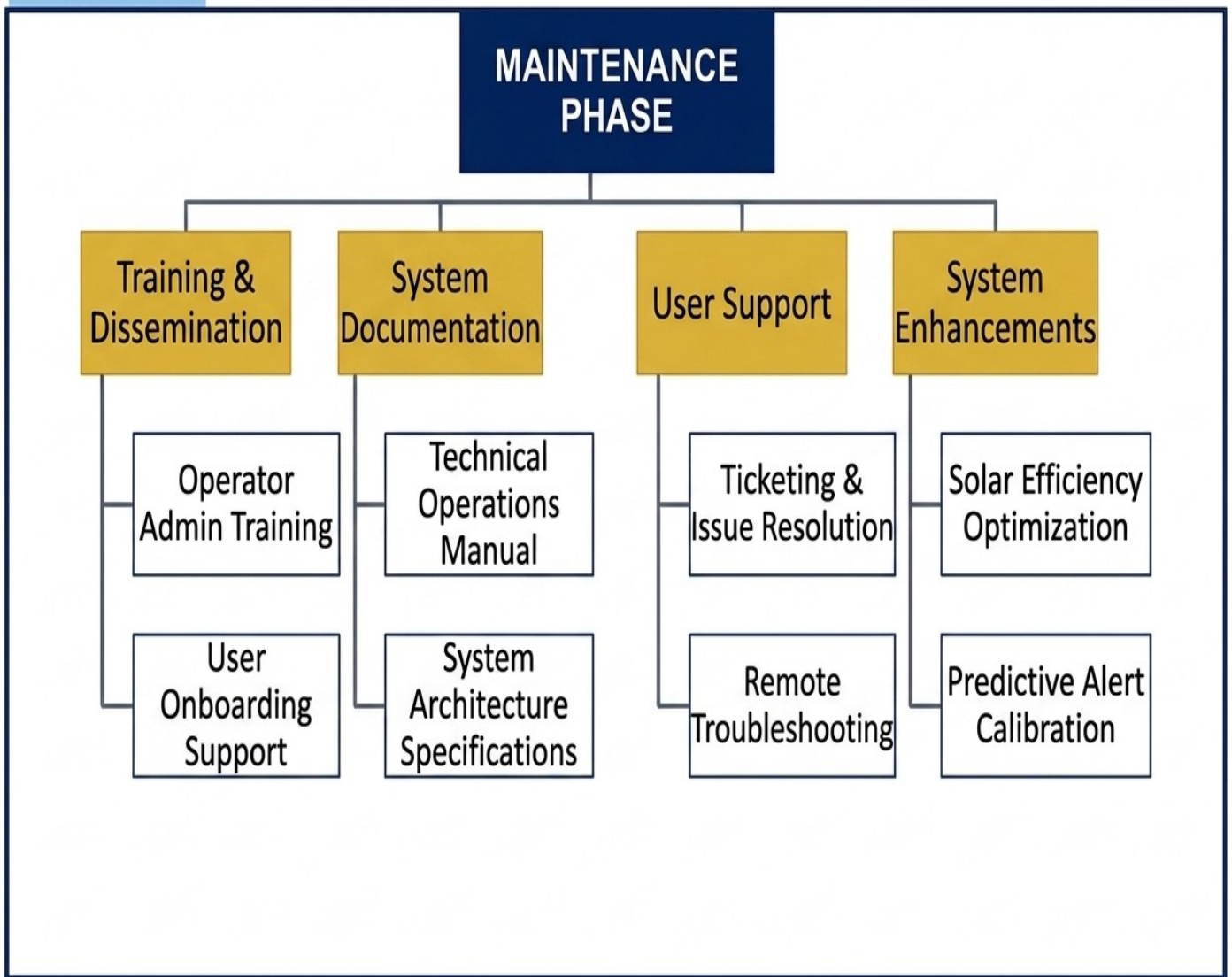


Figure 8. Maintenance Phase WBS

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5.2.2. Schedule allocation

Figure 9 shows the Eco-Link schedule allocation plan designed in Microsoft Excel. The project duration is 135 calendar days with a starting date on August 15, 2023, and expected to end on December 28, 2023. Moreover, Gantt charts and activity lists created in TeamGantt further depict the detailed schedule allocation with milestones and assigned resources for each project phase, including specific IoT hardware procurement and data pipeline testing. Selected TeamGantt screen grabs of individual task management are also presented in the following figures.

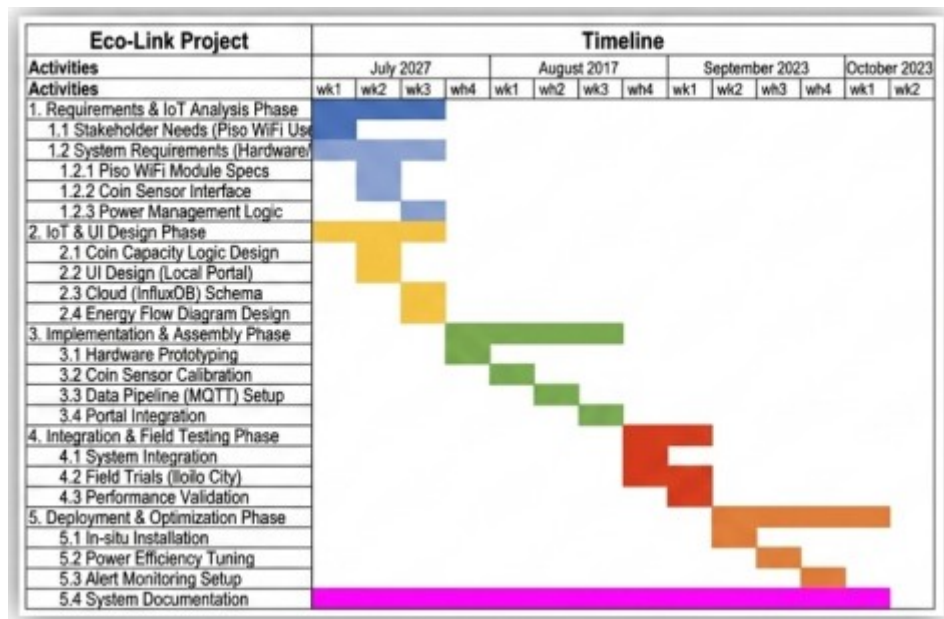


Figure 9. TrainME Gantt Chart

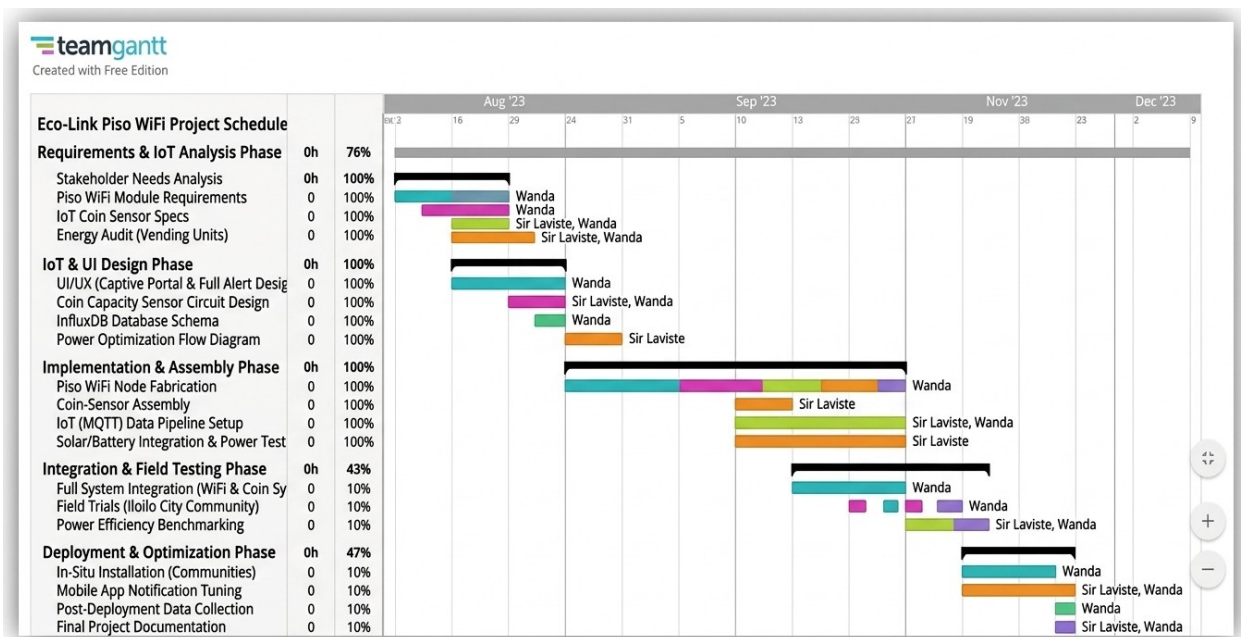


Figure 10. Eco-Link Piso Wifi chart Iot Power Efficiency Data

3/7/2026

TrainME | TeamGantt

Eco-Link Piso WiFi Deployment

Requirements Analysis Phase	% Complete		Start	Due	Assigned
Eco-Link Community Survey & Site Assessment	100%		Oct 2, 2026	Oct 6, 2026	Team Lead
System Performance Requirements (IoT Nodes)	100%		Oct 9, 2026	Oct 11, 2026	Lead Eng
Local Governance Approval (Pilot Site)	100%		Oct 12, 2026	Oct 13, 2026	Project Manage
Energy Resource Inventory	100%		Oct 16, 2026	Oct 17, 2026	Tech Lead
Design Phase	100%		Start	Due	Assigned
IoT Edge Node Circuit Design	100%		Oct 23, 2026	Oct 27, 2026	Eng A
Piso WiFi Mechatronic Integration Design	100%		Oct 30, 2026	Nov 3, 2026	Eng B
Database Schema (Time-Series Telemetry)	100%		Nov 6, 2026	Nov 8, 2026	Data Lead
Power System Diagram (Solar/Battery)	100%		Nov 9, 2026	Nov 10, 2026	Lead Eng
Implementation Phase	100%		Start	Due	Assigned
Edge Node Programming (ESP32)	100%		Nov 13, 2026	Nov 24, 2026	Eng A
Mechatronic Fabrication (Node Enclosures)	100%		Nov 27, 2026	Dec 8, 2026	Eng B
Central Data Pipeline Setup (Cloud)	100%		Nov 27, 2026	Dec 8, 2026	Data Lead
Pilot Network Infrastructure Deployment	100%		Dec 11, 2026	Dec 22, 2023	Field Team
Verification Phase	100%		Start	Due	Assigned
Individual Component Function Testing	100%		Jan 2, 2026	Jan 5, 2026	Eng A
System Integration Testing (Lab)	100%		Jan 8, 2026	Jan 12, 2026	Eng B
Field Performance Validation (Pre-Deployment)	100%		Jan 15, 2026	Jan 19, 2026	Lead Eng
Maintenance Phase	43%		Start	Due	Assigned
Pilot Site Training & Operational Handover	0%		Next Week	Jan 6, 2026	Field Team
Final System Documentation & User Guides	50%		Jan 22, 2026	Feb 9, 2026	Lead Eng
Community Performance Monitoring (Live)	0%		Feb 12, 2026	Feb 23, 2026	Field Team
Network Optimization & Feature Enhancements	0%		Feb 26, 2026	Mar 8, 2026	Tech Lead

200%

Figure 11. Task List with Start and Due Dates

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User Requirements Analysis (Eco-Link Piso WiFi) Sir Levante Wanda Done

% Complete: **100%** Start: **Mar 2, 2026** Due: **Mar 15, 2026** [View in Send](#)

Documents:

EcoLink_Reqs.docx
 Eco-Link specific requirements document.
 by Meta-data updated for 2026
[Collect new version](#)

[Add File](#)

Comments:

- Eco-Link Community Survey & Site Assessment:**
 - Develop site-specific assessment forms and community demand surveys to evaluate pilot areas in Iloilo City.
- IoT Network Requirements Analysis:**
 - Define bandwidth, latency, and node capacity requirements for the edge node mesh network.
- Mechatronic Design for Piso WiFi Vending:**
 - Develop specifications for coin acceptor logic, mechatronic error-coding, and coin capacity alert sensors.
- Power System Analysis (Solar/Battery):**
 - Conduct an energy audit of vending units and specify minimum solar array and battery storage capacity.
- User Interface & Experience (UI/UX) Requirements:**
 - Define constraints for the captive portal, full-screen mobile notifications, and multi-user login workflows.
- Stakeholder Engagement Protocols:**
 - Create standard operating procedures (SOPs) for local governance approval and user demographic profiling.

Weeks: **Days:**
Group: **Requirements Analysis Place**
Type: **Task**
Color:

Dependencies:
 Depends on:
[System Design Phase](#)
[IoT Component Procurement](#)
[Solar Array Fabrication](#)

Video Tip: Watch video to learn about dependencies:
[30 Hide Tip](#)

Figure 12. User Requirements Analysis Task Info

System Requirements Analysis - Research Version Wanda Done

% Complete: **100%** Start: **Aug 1, 2023** End: **Sep 15, 2023** [View in Credit](#)

Documents:

Technical_Requirements_Framework.pdf
 Description of the comprehensive technical framework.
 by Wanda, Aug 2, 2023
[Collect new version](#)

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Technical Stack & Tools:

Data processing & logic Sensor data messaging Time-series data storage

Comments:

Piso WiFi Mechatronic System Integration (Logic for alerts):

- Specific coin acceptor logic for capacity monitoring.
- Integration with a full-screen mobile interface.
- Dynamic multi-user login workflows.

Edge Node Design Constraints:

- Specific power consumption targets.
- Form factor limitation.

Data Transmission & Telemetry Protocols:

- Required message structures and frequency.
- Encryption requirements.

Backend & Power Analytics Requirements:

- Power data processing logic.
- Dashboard metric requirements.

Weeks: **Days:**
Group: **Requirements Analysis Phase**
Type: **Task**
Color:

Dependencies:
[Eco-Link User Demographics Survey](#)
[Site-specific Vending Data Collection](#)
 Dependent on this task

Video Tip: Watch video to learn about dependencies.
[Hide Tip](#)

Figure 13. System Requirements Analysis Task Info

Capstone Project
 Training Management Endeavors (TrainME):
 Web-based Training Needs Assessment and

Prototyping Task Info (Eco-Link Piso WiFi)

Wanda

Done

% Complete: 40%

Start: Apr 1, 2026

End: May 15, 2026

View in Credit

Documents:

Prototyping_Design_V4.dwg

Detailed mechatronic design for coin sensor integration and solar controller.

by Wanda, Apr 3, 2026

Collect new version

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Prototyping Tools & Components:

Multimeter

Circuit testing

Soldering Iron

Node assembly

Piso Module

Coin Logic Board

Solar Array Controller

Power Logic

Comments:

- Coin Capacity Sensor Assembly & Calibration:**
 - Final mounting for specific coin acceptor models.
 - Test procedures with calibrated coin weights.
- Solar Array Integration:**
 - Physical wiring to custom battery packs.
 - Test dynamic power transfer efficiency.
- Local Captive Portal & Mobile Notification Integration:**
 - Implement full-screen alert logic.
 - UX field-trials for notification frequency.
- Network-Setup (MQTT Broker & Database):**
 - Local MQTT broker for device messaging.
 - InfluxDB setup for time-series telemetry.

Weeks: 6

Days: 45

Group: Design Phase

Type: Task

Color: Light Green

Dependencies:

Depends on:

[Mechatronic Design Diagram](#)
[Eco-Link Piso WiFi Mechatronic System Integration](#)

Dependent on this task

[System Integration Testing](#)
[Eco-Link Community Pilot Data Collection](#)

Video Tip: Watch video to learn about dependencies.

Hide Tip

Figure 14. Prototyping Task Info

Database Design Task Info (Eco-Link Piso WiFi)

Wanda

Done

% Complete: 0%

Start: Jul 1, 2026

End: Jul 15, 2026

View in Credit

Documents:

EcoLink_DB_Schema_V1.1.sql

Comprehensive database design schema (normalized) for user data and system telemetry.

by Wanda, Jul 3, 2026

Collect new version

Add File

Database Tools & Software:

Python

Data manipulation

InfluxDB

Time-series data

SQL console

Query formulation

MQTT

Device messaging

Comments:

- Define Normalized Relational Schema for Core Data:**
 - Identify distinct entities (Users, Vending Nodes, Coins, Notifications).
 - Create relational tables with proper primary and foreign keys.
- Integrate Time-Series Power Telemetry (InfluxDB):**
 - Design buckets for power consumption and generation data.
 - Define retention policies and downsampling protocols.
- Establish Secure Device Communication (MQTT):**
 - Implement data broker logic and topic structures for real-time monitoring.
 - Define payload formats for sensor data and alerts.
- Schema Indexing and Database Tuning:**
 - Plan database indexing for critical query paths (e.g., user search, power history).
 - Define query performance benchmarks.

Weeks: 2

Days: 14

Group: Design Phase

Type: Task

Color: Light Green

Dependencies:

Depends on:

[Eco-Link piso-wifi Community Survey](#)
[Mechatronic Design Diagram](#)

Dependent on this task

[IoT Network Requirements Analysis](#)
[Prototyping Phase](#)

Video Tip: Watch video to learn about dependencies.

Hide Tip

Capstone Project

Training Management Endeavors (TrainME):

Web-based Training Needs Assessment and

Figure 15. Database Design Task Info

Capstone Project
Training Management Endeavors (TrainME):
Web-based Training Needs Assessment and

Front-End Implementation Task Info (Eco-Link Piso WiFi)

Wanda

Done

% Complete: 25%

Start: Aug 1, 2026

End: Sep 1, 2026

[View in Credit](#)

Documents:

EcoLink_CaptivePortal_Design_V2.pdf

Final wireframes for user demographics profiling and login interface.
by Wanda, Jul 28, 2026
[Collect new version](#)

Add File

EcoLink_Portal_Wireframe.png

Front-End Stack & Design Tools:

Figma UI/UX Design

VS Code Development

Git Version Control

Chrome DevTools Interface Testing

Comments:

1. Develop Responsive Captive Portal Landing Page:

- Create multi-device compatible layout using HTML/CSS/JS.
- Implement user registration form for required demographics profiling.

2. Code Dynamic Login and User Authentication Flow:

- Build secure data submission logic to interface with database.
- Design multi-user concurrent login workflows with concurrent session management.

3. Integrate Mechatronic Vending Alerts (Full-Screen Mobile):

- Design notification overlay logic for full-screen alerts.
- Establish real-time alert messaging interface (e.g., WebSocket/MQTT to front-end).

4. Implement Site-specific Survey Interface:

- Code survey forms to collect community demand data directly via portal.
- Design survey results dashboard.

Weeks: 4

Days: 31

Group: Implementation Phase

Type: Task

Color: Light Green

Dependencies:

Depends on:

Mechatronic Design Diagram

Eco-Link Piso WiFi Mechatronic System Integration

Dependent on this task

System Integration Testing

Field Trials Protocols

Video Tip: Watch video to learn about dependencies.



[Hide Tip](#)

Figure 16. Front-End Implementation Task Info

Back-End Implementation Task Info (Eco-Link Piso WiFi)

Wanda

Done

% Complete: 30%

Start: Aug 5, 2026

End: Sep 10, 2026

[View in Credit](#)

Documents:

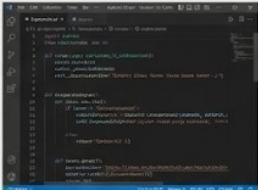
EcoLink_DB_Schema_V3.pdf

PisoLogic_ESP32_Codebase.zip

Backend_API_Documentation.docx replaces Front-end design V2

[Collect new version](#)

Add File

EcoLink_Portal_Wireframe.png

Back-End Stack & Tools:

Python (for logic and processing)

MQTT (Sensor data messaging)

InfluxDB (Time-series data storage)

Docker (Backend containerization)

Comments:

1. Core Database Schema & Data Integrity:

- Design relational tables with specific coin acceptor mechatronic data fields.
- Implement robust input validation for coin denominations and full-screen mobile notifications data.

2. ESP32 & Mechatronic Integration Logic:

- Develop and debug mechatronic logic (C++/MicroPython) for real-time coin sensor capacity alerts.
- Establish secure device-to-backend communication for MQTT telemetry.

3. API Endpoints & Vending Flows:

- Create secure RESTful API endpoints for user demographic data retrieval and site-specific survey input.
- Code backend vending flows for dynamic multi-user concurrent login with mechatronic error-coding.

4. Data Analytics & Dashboard Logic:

- Implement logic for time-series data storage and analysis from Piso WiFi nodes.
- Establish data pipelines from sensors dashboard with community-based surveys.

Weeks: 5

Days: 37

Group: Implementation Phase

Type: Task

Color: Light Green

Dependencies:

Depends on:

Mechatronic Design Diagram

Software Engineering Requirements Analysis

User Requirements Analysis

Dependent on this task

Front-End Implementation Task Info

System Integration Testing

Eco-Link Community Pilot Data Collection

Video Tip:



[Hide Tip](#)

Figure 17. Back-End Implementation Task Info

5.2.3.

Resource allocation

Table 7 shows the relationship between the tasks and resources needed along with the tools required. Table 8 reflects the main and supplemental resources needed for each project phase.

Table 7. Tasks-Resources-Tools Allocation Plan

Task ID	Task Description	Assigned Resource(s)	Required Tools / Equipment
1.1	Stakeholder Needs & Site Assessment	Team Lead, Field Team	Google Forms, Community Survey Sheets
1.2	System Requirements Analysis	Lead Engineer, Tech Lead	VS Code, Technical Documentation Templates
2.1	Prototyping & Circuit Design	Eng A, Eng B	Fritzing, Multimeter, Soldering Station
2.2	UI/UX Captive Portal Design	Eng A, Team Lead	Figma, Web Browsers (Chrome DevTools)

Capstone Project

Training Management Endeavors (TrainME):

Web-based Training Needs Assessment and

2.3	Database Design & Schema Layout	Data Lead	InfluxDB Console, MySQL Workbench
3.1	Front-End Implementation	Eng A	HTML5/CSS3, JavaScript, VS Code, Git
3.2	Back-End API & ESP32 Programming	Eng B, Data Lead	C++/ MicroPython, MQTT Broker (Mosquitto)
3.3	Hardware Fabrication & Assembly	Eng B, Field Team	3D Printer/Acrylic Cutter, Hand Tools
4.1	System Integration Testing	Eng A, Eng B, Tech Lead	Lab Oscilloscope, MQTT-FX Client
4.2	Field Trials (Iloilo City Pilot)	Field Team, Project Manager	Test Coins, Portable Power Monitor
5.1	Deployment & In-Situ Installation	Field Team, Tech Lead	Mounting Brackets, Solar Charge Controller

Capstone Project

Training Management Endeavors (TrainME):

Web-based Training Needs Assessment and

5.2	Monitoring & Documentation	Lead Eng, Team Lead	Microsoft Office, GitHub Documentation
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Capstone Project
Training Management Endeavors (TrainME):
Web-based Training Needs Assessment and
Table 8. Resource Allocation Plan

Project Phase	Main Resources (Primary)	Supplemental Resources (Support)
1. Requirements Analysis Phase	Team Lead Lead Engineer	Project Manager Tech Lead Field Team
2. Design Phase	Eng A (Hardware/UI) Eng B (Mechatronics) Data Lead	Tech Lead Team Lead
3. Implementation Phase	Eng A (Front-End) Eng B (Embedded Software)	Field Team (Physical Assembly) Tech Lead

	Data Lead (Cloud Infrastructure)	
4. Verification & Field Testing Phase	Eng A & Eng B	Project Manager
	Field Team	Lead Engineer
5. Deployment & Maintenance Phase	Field Team	Team Lead
	Tech Lead	Lead Engineer

5.2.4.

Budget allocation

This table details the hardware components, operational expenses, and software subscriptions required to build and deploy the Eco-Link prototype.

Table 9: Itemized Project Cost Breakdown

Categor y	Item Description	Qty	Unit Cost (PHP)	Total Cost (PHP)
Hardwa re Compo nents	Custom Heavy-Duty Coin Slot (Multi-coin)	● 2 units	₱1,200.00	₱2,400.00

Capstone Project

Training Management Endeavors (TrainME):

Web-based Training Needs Assessment and

	ESP32 Wi-Fi/Bluetooth Microcontroller Node	● 3 units	₱450.00	₱1,350.00
	Infrared/Optical Coin Capacity Sensors	● 4 units	₱250.00	₱1,000.00
	50W Solar Panel & 12V Charge Controller	1 set	₱4,500.00	₱4,500.00
	12V Deep Cycle Lead-Acid Battery (Back-up)	1 unit	₱3,200.00	₱3,200.00
	Mechatronic Enclosure (Acrylic & Marine Plywood)	1 lot	₱2,500.00	₱2,500.00
Software & Cloud	InfluxDB Cloud Storage Subscription (Tear-1)	5 mos.	₱850.00	₱4,250.00
	Twilio API Gateway SMS Notification Credits	1 lot	₱1,500.00	₱1,500.00
Operational & Field	Field Trial Travel & Transportation (Iloilo City)	1 lot	₱3,500.00	₱3,500.00

Capstone Project*Training Management Endeavors (TrainME):**Web-based Training Needs Assessment and*

	Miscellaneous Tools (Soldering, Wires, Fasteners)	1 lot	₱1,800.00	₱1,800.00
Contingency	Emergency Component Replacement Fund (10%)	1 lot	₱2,600.00	₱2,600.00
TOTAL				₱28,600.00

Capstone Project

Training Management Endeavors (TrainME):

Web-based Training Needs Assessment and

Table 10: Budget Allocation per Project Phase

Phase ID	Project Phase	Allocation (%)	Estimated Cost (PHP)	Primary Cost Drivers
1.0	Requirements & IoT Analysis Phase	5%	₱1,430.00	Stakeholder survey printing, initial site assessments.
2.0	IoT & UI Design Phase	10%	₱2,860.00	Design tools, structural schematics, and UI prototyping.
3.0	Implementation & Assembly Phase	55%	₱15,730.00	Procurement of primary hardware, solar kits, and fabrication materials.
4.0	Integration & Field Testing Phase	20%	₱5,720.00	Cloud subscriptions activation, field deployment transport, test coin sets.
5.0	Deployment & Optimization Phase	10%	₱2,860.00	On-site bracket mounting, user manual printing, final handover adjustments.
TOTAL		100%	₱28,600.00	

Capstone Project

Training Management Endeavors (TrainME):

Web-based Training Needs Assessment and

5.3. Control plan

5

5.0 Project Control and Technical Process Plans

This section details the control mechanisms, engineering methodologies, software tools, and validation frameworks used across the lifecycle of the **Eco-Link Piso WiFi** system.

5.1 Project Control Plans

5.1.1 Requirements Control Plan

The Project Manager handles defect tracking, embedded logic package dependencies (ESP32 libraries), and firmware code reviews. A metrics report containing code health, power-consumption footprints, and sensor accuracy is prepared weekly. Hardware/software scope creeping is strictly monitored to ensure that secondary functions do not exceed the power budget allocated by stakeholders.

5.1.2 Schedule Control Plan

Actual start and finish dates are logged in real-time within **TeamGantt**. These metrics are compared weekly against the original baseline to check for milestones slipping (e.g., sensor calibration delays). Resource reallocation, such as shifting team members to mechatronic assembly if fabrication stalls, is decided collaboratively with advisors and stakeholders.

5.1.3 Budget Control Plan

Financial expenditures are strictly itemized in a master Microsoft Excel sheet tracking debits and credits against a baseline ceiling of ₱28,600.00. Receipts are archived for all hardware components (solar array, controllers, sensors). Earned Value Management (EVM) metrics are checked at hardware prototyping and field testing milestones to balance work completed against actual financial burn rates.

5.1.4 Quality Control Plan

Hardware tolerances and telemetry accuracy are verified at the completion of each phase. Sensor validation tests (drop-testing 100 random denominations) are logged to ensure coin acceptance rates remain $\geq 98\%$. Data pipelines must hit a quality ceiling of $\leq 1\%$ MQTT packet drops during continuous transmission tests.

5.1.5 Reporting Plan

- **Progress Meetings:** Held twice a week (Wednesday and Saturday) to review firmware builds and structural assembly progress.
- **Status Reports:** Generated weekly by the Project Manager, tracking completion percentages, power-budget draw profiles, and estimated engineering hours remaining.
- **Critical Alerts:** Major mechatronic anomalies (e.g., battery overcharging or coin jams) are reported immediately.
- **Communication Channels:** Gmail and a dedicated Messenger group serve as the configuration control hub, while Google Drive acts as the central repository for CAD designs, schematics, and source code.

5.1.6 Metrics Collection Plan

Developers deliver stable, error-free firmware compiles to the configuration hub every Wednesday by 11:00 PM. Code changes are pushed to a localized tracking repository by Saturday at 8:00 AM. Every distribution package includes compiled binaries (.bin files for the ESP32), structural 3D CAD files, and an updating requirements trace matrix.

The primary risk is schedule slippage during the mechatronic integration phase, categorized as **High** due to the custom nature of the coin capacity optical array and solar-charging algorithms. Lacking historical data, estimates are built on analytical approximations. If a phase falls behind, project scale modifications (e.g., shifting from web-dashboard analytics to localized SMS alerts) are initiated to protect the deployment deadline.

5.1.8 Project Closeout Plan

All system manuals, electrical schematics, and testing records must be uploaded to the central Google Drive folder 48 hours before the final defense deadline. Structural adjustments, code refactoring, or text edits suggested by advisors must be coordinated 24 hours prior, with a mandatory final sign-off by all team members 12 hours before final submission.

5.2 Technical Process Plans

5.2.1 Process Model

The Eco-Link system uses an **Overlapping Waterfall Model with Feedback Loops and Prototyping**. Design tasks begin mid-way through requirements analysis to model circuit layouts early. Feedback from initial physical testing loops back directly into design alterations, allowing structural updates to happen prior to deployment.

Below is the structured lifecycle model mapping the specific project milestones and engineering artifacts:

Table 11. Methods and Techniques Plan

Category	Methods and Techniques
Requirements	* Document Sampling (Piso WiFi operation logs)
	* Structured Interviews (Local operators & vending patrons)
	* Field Observations (Coin filling cycles, environment constraints)
	* Benchmarking (Power efficiency targets)
Formal Specification & Analysis	* Unified Modeling Language (UML) Use Cases for portal workflows
	* Structural Class & Sequence Diagrams for telemetry messaging protocols

* Circuit Schematics (Fritzing/Altium) for mechatronic routing

Prototyping

* Low-fidelity paper layouts for the captive portal interface

* Functional hardware prototypes (ESP32 breakout boards) to test sensor data scaling

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Table 12. Hardware, Software, and System Tools Plan

Category	Tools
Operating System & Development	Windows 10 / Windows 11, VS Code (PlatformIO extension)
Firmware Environment	C++ (Arduino Framework), Espressif ESP32 Core Core SDK
Database & Analytics Backend	InfluxDB (Time-series power logging), Mosquitto MQTT Broker
UI Design & Prototyping	Figma, TailwindCSS, Vanilla JavaScript, HTML5
Hardware Engineering	Fritzing (PCB design), SolidWorks (Enclosure CAD models)
Tracking & Documentation	TeamGantt, Microsoft Excel, Microsoft Word, GitHub

Export to Sheets

5.3 Supporting Process Plans

5.3.1 Verification and Validation (V&V) Framework

The project uses continuous cross-phase validation via traceability analysis, evaluation loops, physical interface checks, and rigorous hardware/software testing.

Table 13. Eco-Link Verification and Validation Plan

Phase	V&V Input Artifacts	V&V Tasks	V&V Output Artifacts
Requirements	Preliminary Scope Document, Field Survey Data	* Traceability validation of stakeholder power budgets	Approved Software/Hardware Requirements Specification (SRS)

Capstone Project*Training Management Endeavors (TrainME):**Web-based Training Needs Assessment and*

		* Input/Output interface boundary analysis	
Design	SRS, Component Datasheets, Pinout Maps	* Architectural evaluation of solar charge regulators	Schematics, Database Schema Layout, Physical Test Plan
		* Telemetry database schema verification	
		* Test-case design generation	
Implementation	Embedded C++ Code, Fabricated Enclosure	* Code analysis for memory leaks on ESP32	Verified Sub-modules, Initial Calibration Logs
		* Physical dimension checks against CAD blueprints	
		* Unit execution of sensor reading loops	
System Test	Integrated Vending Unit, Firmware Binaries	* Black Box stress testing of coin drop sequences	Comprehensive Test Log, System Anomaly Reports
		* White Box code pathway coverage analysis	
		* Captive portal stress load testing	
Installation	Production Build, Operational Deployment Site	* Site power configuration audit	V&V Final Summary Report, System Handover Manifest

Capstone Project

Training Management Endeavors (TrainME):

Web-based Training Needs Assessment and

* Final validation of
remote over-the-air
SMS alerts

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5.3.2 Research Documentation Portfolio

The following project deliverables are generated, updated, and archived across the system lifecycle:

1. **Software Project Management Plan (SPMP):** Scheduling, financial allocation, and control strategies.
2. **Software Requirements Specification (SRS):** Detailed hardware constraints, network latency ceilings, and interface requirements.
3. **System Architecture and Design Description:** Circuit schematics, mechatronic 3D CAD blueprints, and database ERDs.
4. **Verification and Validation Test Description:** Comprehensive unit, integration, and user-acceptance test configurations and operational results files.